

SNES CPU-visible register map

Every register the 65816 can reach directly (`$2100–$21FF` , `$4016–$43FF`), generated from the annotated HAL headers in `platforms/snes/snes_*.h` — the single source of truth. The audio processor (SPC700 + S-DSP) is reachable only through the four APU ports below and its internal registers are a separate address space (out of scope here; see the [hardware summary](#)). For the CPU core itself see the [65816 reference](#).

Bit-field facts are per nocash *fullsnes*, the SNESdev wiki, and anomie's docs ([SOURCES.md](#)).

PPU — Picture Processing Unit (\$2100–\$213F)

Display, backgrounds, OAM, VRAM, Mode 7, CGRAM, windows, colour math, and the read ports. — `snes_ppu.h` , 66 registers.

Addr	Name	R/W	Purpose
\$2100	REG_INIDISP	W	Display control 1 — force-blank + master brightness.
\$2101	REG_OBSEL	W	Object (sprite) size + tile-base select.
\$2102	REG_OAMADDL	W	OAM word address, low byte (0..255).
\$2103	REG_OAMADDH	W	OAM address high + priority rotation.
\$2104	REG_OAMDATA	W	OAM data write (auto-increments OAMADD; write low then high).
\$2105	REG_BGMODE	W	BG mode + per-layer tile size.
\$2106	REG_MOSAIC	W	Mosaic block size + per-BG enable.
\$2107	REG_BG1SC	W	BG1 tilemap base + map size.
\$2108	REG_BG2SC	W	BG2 tilemap base + size (see BG1SC).
\$2109	REG_BG3SC	W	BG3 tilemap base + size (see BG1SC).
\$210A	REG_BG4SC	W	BG4 tilemap base + size (see BG1SC).
\$210B	REG_BG12NBA	W	BG1/BG2 character (tile) data base.

Addr	Name	R/W	Purpose
\$210C	REG_BG34NBA	W	BG3/BG4 character data base.
\$210D	REG_BG1H0FS	W	BG1 horizontal scroll (write twice; 10-bit, +Mode7 share).
\$210E	REG_BG1V0FS	W	BG1 vertical scroll (write twice; 10-bit).
\$210F	REG_BG2H0FS	W	BG2 horizontal scroll (write twice).
\$2110	REG_BG2V0FS	W	BG2 vertical scroll (write twice).
\$2111	REG_BG3H0FS	W	BG3 horizontal scroll (write twice).
\$2112	REG_BG3V0FS	W	BG3 vertical scroll (write twice).
\$2113	REG_BG4H0FS	W	BG4 horizontal scroll (write twice).
\$2114	REG_BG4V0FS	W	BG4 vertical scroll (write twice).
\$2115	REG_VMAIN	W	VRAM address increment control.
\$2116	REG_VMADDL	W	VRAM word address, low byte.
\$2117	REG_VMADDH	W	VRAM word address, high byte.
\$2118	REG_VMDATAL	W	VRAM data, low byte (increments per VMAIN if INCHIGH=0).
\$2119	REG_VMDATAH	W	VRAM data, high byte (increments per VMAIN if INCHIGH=1).
\$2116	REG_VMADD ¹⁶	W	VRAM word address, 16-bit access alias over VMADDL/H.
\$2118	REG_VMDATA ¹⁶	W	VRAM data, 16-bit access alias over VMDATAL/H (low then high).
\$211A	REG_M7SEL	W	Mode 7 screen settings.
\$211B	REG_M7A	W	Mode 7 matrix A (signed 8.8; write low,high). Also hw-multiply factor A.
\$211C	REG_M7B	W	Mode 7 matrix B (signed 8.8; write low,high). Also hw-multiply factor B (8-bit).
\$211D	REG_M7C	W	Mode 7 matrix C (signed 8.8; write low,high).

Addr	Name	R/W	Purpose
\$211E	REG_M7D	W	Mode 7 matrix D (signed 8.8; write low,high).
\$211F	REG_M7X	W	Mode 7 center X (signed 13-bit; write twice).
\$2120	REG_M7Y	W	Mode 7 center Y (signed 13-bit; write twice).
\$2121	REG_CGADD	W	CGRAM (palette) entry address (0..255).
\$2122	REG_CGDATA	W	CGRAM data port — BGR555, write low byte then high byte.
\$2123	REG_W12SEL	W	Window mask settings for BG1/BG2 (2 bits per window/layer).
\$2124	REG_W34SEL	W	Window mask settings for BG3/BG4.
\$2125	REG_WOBJSEL	W	Window mask settings for OBJ + color-math.
\$2126	REG_WH0	W	Window 1 left position (pixel 0..255).
\$2127	REG_WH1	W	Window 1 right position.
\$2128	REG_WH2	W	Window 2 left position.
\$2129	REG_WH3	W	Window 2 right position.
\$212A	REG_WBGLOG	W	Window-1/2 combine logic per BG (00 OR, 01 AND, 10 XOR, 11 XNOR).
\$212B	REG_WOBJLOG	W	Window combine logic for OBJ + color-math.
\$212C	REG_TM	W	Main-screen layer enable.
\$212D	REG_TS	W	Sub-screen layer enable (bits as TM).
\$212E	REG_TMW	W	Main-screen window-mask enable per layer (bits as TM).
\$212F	REG_TSW	W	Sub-screen window-mask enable per layer (bits as TM).
\$2130	REG_CGWSEL	W	Color-math control window + source.
\$2131	REG_CGADSUB	W	Color-math operation + per-layer enable.
\$2132	REG_COLDATA	W	Fixed color for color math (write once per channel).

Addr	Name	R/W	Purpose
\$2133	REG_SETINI	W	Display mode / interlace settings.
\$2134	REG_MPYL	R	Mode 7 / hw multiply result, low byte (M7A signed16 x M7B signed8 = signed24).
\$2135	REG_MPYM	R	multiply result, middle byte.
\$2136	REG_MPYH	R	multiply result, high byte.
\$2137	REG_SLHV	R	software latch: a dummy read latches the H/V counters into OPHCT/OPVCT.
\$2138	REG_OAMDATAREAD	R	OAM data read (auto-increments OAMADD).
\$2139	REG_VMDATALREAD	R	VRAM data read, low byte (increments per VMAIN).
\$213A	REG_VMDATAHREAD	R	VRAM data read, high byte.
\$213B	REG_CGDATAREAD	R	CGRAM data read (low then high).
\$213C	REG_OPHCT	R	horizontal counter latch (read twice: low then high bit 8; 0..339).
\$213D	REG_OPVCT	R	vertical counter latch (read twice; 0..261 NTSC).
\$213E	REG_STAT77	R	PPU1 (5C77) status.
\$213F	REG_STAT78	R	PPU2 (5C78) status.

► Bit fields

¹⁶ = 16-bit access alias over the adjacent L/H byte pair.

DMA / HDMA (\$4300–\$437F, + \$420B/\$420C)

Eight transfer channels plus the MDMAEN/HDMAEN enable triggers. — `snescdma.h`, 90 registers.

Addr	Name	R/W	Purpose
\$420B	REG_MDMAEN	W	General-purpose DMA enable + trigger; CPU stalls until done.
\$420C	REG_HDMAEN	W	HDMA enable (per-scanline); bit n = arm channel n for HDMA.

Addr	Name	R/W	Purpose
\$4300	REG_DMAP0	RW	Channel 0 transfer parameters.
\$4301	REG_BBAD0	RW	Channel 0 B-bus destination (low byte of \$21xx, e.g. \$18 -> VMDATA).
\$4302	REG_A1T0L	RW	Channel 0 A-bus source address, low byte.
\$4303	REG_A1T0H	RW	Channel 0 A-bus source address, high byte.
\$4304	REG_A1B0	RW	Channel 0 A-bus source bank.
\$4305	REG_DAS0L	RW	Channel 0 byte count low (GP-DMA) / HDMA indirect addr low. 0 count = 65536.
\$4306	REG_DAS0H	RW	Channel 0 byte count high / HDMA indirect addr high.
\$4307	REG_DASB0	RW	Channel 0 HDMA indirect address bank (HDMA only).
\$4308	REG_A2A0L	RW	Channel 0 HDMA table current address, low (HDMA only).
\$4309	REG_A2A0H	RW	Channel 0 HDMA table current address, high (HDMA only).
\$430A	REG_NTRL0	RW	Channel 0 HDMA line counter (HDMA only).
\$4310	REG_DMAP1	RW	Channel 1 transfer parameters (see DMAP0).
\$4311	REG_BBAD1	RW	Channel 1 B-bus destination.
\$4312	REG_A1T1L	RW	Channel 1 A-bus source low.
\$4313	REG_A1T1H	RW	Channel 1 A-bus source high.
\$4314	REG_A1B1	RW	Channel 1 A-bus source bank.
\$4315	REG_DAS1L	RW	Channel 1 byte count low / HDMA indirect low.
\$4316	REG_DAS1H	RW	Channel 1 byte count high / HDMA indirect high.
\$4317	REG_DASB1	RW	Channel 1 HDMA indirect bank.
\$4318	REG_A2A1L	RW	Channel 1 HDMA table address low.
\$4319	REG_A2A1H	RW	Channel 1 HDMA table address high.

Addr	Name	R/W	Purpose
\$431A	REG_NTRL1	RW	Channel 1 HDMA line counter.
\$4320	REG_DMAP2	RW	Channel 2 transfer parameters (see DMAP0).
\$4321	REG_BBAD2	RW	Channel 2 B-bus destination.
\$4322	REG_A1T2L	RW	Channel 2 A-bus source low.
\$4323	REG_A1T2H	RW	Channel 2 A-bus source high.
\$4324	REG_A1B2	RW	Channel 2 A-bus source bank.
\$4325	REG_DAS2L	RW	Channel 2 byte count low / HDMA indirect low.
\$4326	REG_DAS2H	RW	Channel 2 byte count high / HDMA indirect high.
\$4327	REG_DASB2	RW	Channel 2 HDMA indirect bank.
\$4328	REG_A2A2L	RW	Channel 2 HDMA table address low.
\$4329	REG_A2A2H	RW	Channel 2 HDMA table address high.
\$432A	REG_NTRL2	RW	Channel 2 HDMA line counter.
\$4330	REG_DMAP3	RW	Channel 3 transfer parameters (see DMAP0).
\$4331	REG_BBAD3	RW	Channel 3 B-bus destination.
\$4332	REG_A1T3L	RW	Channel 3 A-bus source low.
\$4333	REG_A1T3H	RW	Channel 3 A-bus source high.
\$4334	REG_A1B3	RW	Channel 3 A-bus source bank.
\$4335	REG_DAS3L	RW	Channel 3 byte count low / HDMA indirect low.
\$4336	REG_DAS3H	RW	Channel 3 byte count high / HDMA indirect high.
\$4337	REG_DASB3	RW	Channel 3 HDMA indirect bank.
\$4338	REG_A2A3L	RW	Channel 3 HDMA table address low.
\$4339	REG_A2A3H	RW	Channel 3 HDMA table address high.

Addr	Name	R/W	Purpose
\$433A	REG_NTRL3	RW	Channel 3 HDMA line counter.
\$4340	REG_DMAP4	RW	Channel 4 transfer parameters (see DMAP0).
\$4341	REG_BBAD4	RW	Channel 4 B-bus destination.
\$4342	REG_A1T4L	RW	Channel 4 A-bus source low.
\$4343	REG_A1T4H	RW	Channel 4 A-bus source high.
\$4344	REG_A1B4	RW	Channel 4 A-bus source bank.
\$4345	REG_DAS4L	RW	Channel 4 byte count low / HDMA indirect low.
\$4346	REG_DAS4H	RW	Channel 4 byte count high / HDMA indirect high.
\$4347	REG_DASB4	RW	Channel 4 HDMA indirect bank.
\$4348	REG_A2A4L	RW	Channel 4 HDMA table address low.
\$4349	REG_A2A4H	RW	Channel 4 HDMA table address high.
\$434A	REG_NTRL4	RW	Channel 4 HDMA line counter.
\$4350	REG_DMAP5	RW	Channel 5 transfer parameters (see DMAP0).
\$4351	REG_BBAD5	RW	Channel 5 B-bus destination.
\$4352	REG_A1T5L	RW	Channel 5 A-bus source low.
\$4353	REG_A1T5H	RW	Channel 5 A-bus source high.
\$4354	REG_A1B5	RW	Channel 5 A-bus source bank.
\$4355	REG_DAS5L	RW	Channel 5 byte count low / HDMA indirect low.
\$4356	REG_DAS5H	RW	Channel 5 byte count high / HDMA indirect high.
\$4357	REG_DASB5	RW	Channel 5 HDMA indirect bank.
\$4358	REG_A2A5L	RW	Channel 5 HDMA table address low.
\$4359	REG_A2A5H	RW	Channel 5 HDMA table address high.

Addr	Name	R/W	Purpose
\$435A	REG_NTRL5	RW	Channel 5 HDMA line counter.
\$4360	REG_DMAP6	RW	Channel 6 transfer parameters (see DMAP0).
\$4361	REG_BBAD6	RW	Channel 6 B-bus destination.
\$4362	REG_A1T6L	RW	Channel 6 A-bus source low.
\$4363	REG_A1T6H	RW	Channel 6 A-bus source high.
\$4364	REG_A1B6	RW	Channel 6 A-bus source bank.
\$4365	REG_DAS6L	RW	Channel 6 byte count low / HDMA indirect low.
\$4366	REG_DAS6H	RW	Channel 6 byte count high / HDMA indirect high.
\$4367	REG_DASB6	RW	Channel 6 HDMA indirect bank.
\$4368	REG_A2A6L	RW	Channel 6 HDMA table address low.
\$4369	REG_A2A6H	RW	Channel 6 HDMA table address high.
\$436A	REG_NTRL6	RW	Channel 6 HDMA line counter.
\$4370	REG_DMAP7	RW	Channel 7 transfer parameters (see DMAP0).
\$4371	REG_BBAD7	RW	Channel 7 B-bus destination.
\$4372	REG_A1T7L	RW	Channel 7 A-bus source low.
\$4373	REG_A1T7H	RW	Channel 7 A-bus source high.
\$4374	REG_A1B7	RW	Channel 7 A-bus source bank.
\$4375	REG_DAS7L	RW	Channel 7 byte count low / HDMA indirect low.
\$4376	REG_DAS7H	RW	Channel 7 byte count high / HDMA indirect high.
\$4377	REG_DASB7	RW	Channel 7 HDMA indirect bank.
\$4378	REG_A2A7L	RW	Channel 7 HDMA table address low.
\$4379	REG_A2A7H	RW	Channel 7 HDMA table address high.

Addr	Name	R/W	Purpose
\$437A	REG_NTRL7	RW	Channel 7 HDMA line counter.

► Bit fields

CPU I/O — Ricoh 5A22 (\$4200–\$4217)

Interrupt control, the hardware multiply/divide unit, the H/V IRQ timers, and FastROM select. — `snescpu.h`, 25 registers.

Addr	Name	R/W	Purpose
\$4200	REG_NMITIMEN	W	Interrupt + auto-joypad enable.
\$4201	REG_WRIO	W	Programmable I/O port (output). Bit 7 drives the controller-port
\$4202	REG_WRPYA	W	Unsigned multiply: factor A (8-bit).
\$4203	REG_WRPYB	W	Unsigned multiply: factor B (8-bit); writing starts 8x8->16, result in RDMPY after ~8 cycles.
\$4204	REG_WRDIVL	W	Unsigned divide: dividend low (16-bit dividend).
\$4205	REG_WRDIVH	W	Unsigned divide: dividend high.
\$4206	REG_WRDIVB	W	Unsigned divide: divisor (8-bit); writing starts 16/8, quotient in RDDIV / remainder in RDMPY after ~16 cycles.
\$4204	REG_WRDIV ¹⁶	W	16-bit access alias for the divide dividend (WRDIVL/H).
\$4207	REG_HTIMEL	W	H-count IRQ position, low (0..339).
\$4208	REG_HTIMEH	W	H-count IRQ position, high (bit 0).
\$4209	REG_VTIMEL	W	V-count IRQ position, low (0..261 NTSC).
\$420A	REG_VTIMEH	W	V-count IRQ position, high (bit 0).
\$4207	REG_HTIME ¹⁶	W	16-bit access alias for the H IRQ position (HTIMEL/H).
\$4209	REG_VTIME ¹⁶	W	16-bit access alias for the V IRQ position (VTIMEL/H).
\$420D	REG_MEMSEL	W	ROM access speed.

Addr	Name	R/W	Purpose
\$4210	REG_RDNMI	R	NMI flag + CPU version.
\$4211	REG_TIMEUP	R	IRQ flag.
\$4212	REG_HVBJOY	R	Blanking + auto-joypad status.
\$4213	REG_RDIO	R	Programmable I/O port (input).
\$4214	REG_RDDIVL	R	Divide quotient, low.
\$4215	REG_RDDIVH	R	Divide quotient, high.
\$4216	REG_RDMPYL	R	Multiply product low (or divide remainder low).
\$4217	REG_RDMPYH	R	Multiply product high (or divide remainder high).
\$4214	REG_RDDIV ¹⁶	R	16-bit access alias for the divide quotient (RDDIVL/H).
\$4216	REG_RDMPY ¹⁶	R	16-bit access alias for the multiply product / remainder (RDMPYL/H).

► Bit fields

¹⁶ = 16-bit access alias over the adjacent L/H byte pair.

Controllers / Joypad (\$4016–\$4017, \$4218–\$421F)

Serial controller access and the automatic-read pad latches. — `snesc_joypad.h` , 14 registers.

Addr	Name	R/W	Purpose
\$4016	REG_JOYSER0	RW	Write: strobe/latch both ports. Read: controller 1 serial bit.
\$4017	REG_JOYSER1	R	Controller 2 serial bit (read).
\$4218	REG_JOY1L	R	Controller 1, low byte.
\$4219	REG_JOY1H	R	Controller 1, high byte.
\$421A	REG_JOY2L	R	Controller 2, low byte (layout as JOY1L).
\$421B	REG_JOY2H	R	Controller 2, high byte (layout as JOY1H).

Addr	Name	R/W	Purpose
\$421C	REG_JOY3L	R	Controller 3 (multitap), low byte.
\$421D	REG_JOY3H	R	Controller 3 (multitap), high byte.
\$421E	REG_JOY4L	R	Controller 4 (multitap), low byte.
\$421F	REG_JOY4H	R	Controller 4 (multitap), high byte.
\$4218	REG_JOY1 ¹⁶	R	Controller 1, full 16-bit button word.
\$421A	REG_JOY2 ¹⁶	R	Controller 2, 16-bit button word (layout as JOY1).
\$421C	REG_JOY3 ¹⁶	R	Controller 3, 16-bit button word.
\$421E	REG_JOY4 ¹⁶	R	Controller 4, 16-bit button word.

► Bit fields

¹⁶ = 16-bit access alias over the adjacent L/H byte pair.

APU ports — audio mailbox (\$2140–\$2143)

The four ports to the (separate) SPC700/S-DSP audio processor. — `snescpu.h`, 4 registers.

Addr	Name	R/W	Purpose
\$2140	REG_APUI00	RW	APU I/O port 0 (CPU<->SPC700 mailbox; also the command port).
\$2141	REG_APUI01	RW	APU I/O port 1.
\$2142	REG_APUI02	RW	APU I/O port 2.
\$2143	REG_APUI03	RW	APU I/O port 3.

WRAM access port (\$2180–\$2183)

A CPU window into the full 128 KB of work RAM. — `snescpu.h`, 4 registers.

Addr	Name	R/W	Purpose
\$2180	REG_WMDATA	RW	WRAM data port; each access auto-increments WMADD.
\$2181	REG_WMADDL	W	WRAM address, low byte.
\$2182	REG_WMADDM	W	WRAM address, middle byte.
\$2183	REG_WMADDH	W	WRAM address, high byte.

► Bit fields

203 registers across 6 subsystems.